Highlights

Proxy4Tun: Proxy-driven quality control for tunnel-lining segmentation in point clouds

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- Proxy4Tun estimates segmentation accuracy without ground-truth (GT) labels at deployment.
- Proxy calibration uses one labelled subset and one expert reference per lining family.
- Frozen proxy correctly ranks all 27 reference/degraded pairs on unseen scans.
- GT-free self-refinement improves mean mIoU with both LLM and adaptive random-search.
- Retrospective analysis shows refinement gains concentrate at initial GT mIoU of 0.5–0.7.

Proxy4Tun: Proxy-driven quality control for tunnel-lining segmentation in point clouds

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ABSTRACT

Tunnel-lining segmentation supports automated inspection using point clouds, but varying lining geometry, scanning conditions and limited ground-truth (GT) labels complicate accuracy assessment on new scans. Proxy4Tun estimates segmentation accuracy from intermediate pipeline measurements, with calibration using one labelled subset and one expert reference per lining family. Across three Seg2Tunnel lining families, bounded space-filling sampling and uncertainty-guided Bayesian exploration generate 120 calibration runs. Ablation analysis identifies a four-feature Ridge proxy that estimates mean intersection-over-union (mIoU) without deployment GT labels. On 54 runs from 27 held-out subsets, the frozen proxy achieves a Spearman correlation of 0.817 and a mean absolute error of 0.109 while correctly ranking all 27 reference/degraded pairs. On 22 cases with initial proxy scores in [0.5, 0.7], GT-free self-refinement raises mean mIoU from 0.738 to 0.751–0.760 across three separately evaluated large language models and to 0.752 with with adaptive random search, all using the same proxy. Retrospective analysis shows that refinement gains concentrate in cases with initial GT mIoU of 0.5–0.7 and come mainly from reducing whole-ring class swaps. These findings demonstrate that limited development supervision can support deployment quality control (QC) and bounded parameter refinement. We position Proxy4Tun as an exploratory QC framework rather than a deployment-ready tool for verifying segmentation accuracy.

1. Introduction

Regular inspection of segmental tunnel linings supports structural health assessment and long-term operational safety. Large tunnel networks and restricted access make manual inspection resource-intensive, motivating cost-effective data collection and analysis [1, 2, 3]. Laser scanning enables rapid acquisition of lining surfaces as 3D point clouds [4, 5, 6]. Segment-level inspection analyses require assigning the measured points to their corresponding lining segments. However, segmentation accuracy varies with lining geometry, joint layout, occlusion, and scan quality. Manually labelling new scans to assess accuracy and adjusting pipeline parameters impose recurring costs. Practitioners therefore need support in deciding whether to accept an output, attempt refinement, or flag it for human review without new ground-truth (GT) labels (Fig. 1).

Existing segmentation approaches can be compared using two practical deployment questions: (i) how they cope with changing tunnel and scanning conditions, and (ii) what evidence they provide that a specific output is accurate. Engineer-designed geometric pipelines use handcrafted rules and thresholds with interpretable checks, but changes in geometry or scan quality can require manual tuning or redesign [4, 6, 7]. Supervised models can be retrained on labelled data and may provide confidence or uncertainty scores, but the relationship between these scores and segmentation accuracy requires validation before they can support QC on new scans [8, 9]. Foundation-model pipelines can reduce per-project annotation requirements and support adaptation through prompts and preprocessing parameters [10, 11, 12]. However, the ability to adjust a pipeline does not establish whether the resulting segmentation is better or accurate enough to accept.

Previous studies show that segmentation accuracy can be estimated without GT labels for the new case. In medical imaging, quality estimators learn from labelled development data or assess new outputs through labelled reference collections [13, 14, 15]. It remains unclear how well these approaches transfer to tunnel point-cloud pipelines, where annotations are limited and processing depends on engineering reference configurations. This motivates our central research question: *can measurements produced during tunnel segmentation be calibrated using one labelled subset and one expert-tuned reference configuration per lining family, then used to estimate accuracy and guide parameter refinement on unseen scans without new GT labels?*

Proxy4Tun addresses this question by estimating the accuracy of a candidate segmentation from intermediate pipeline measurements. The candidate measurements describe two aspects of processing. Preprocessing quality (PQ) assesses the completeness and geometric plausibility of the representation supplied to the segmentation stage. Structure coherence (SC) assesses agreement between predicted boundaries and detected joints, segmentation coverage, label-order consistency across rings, and agreement between detected and expected ring counts. The final proxy retains joint–boundary correspondence, coverage, and ring-count residual, together with the PQ missing-depth rate.

During development, we learn how these measurements relate to GT mIoU, then freeze the estimator for deployment.

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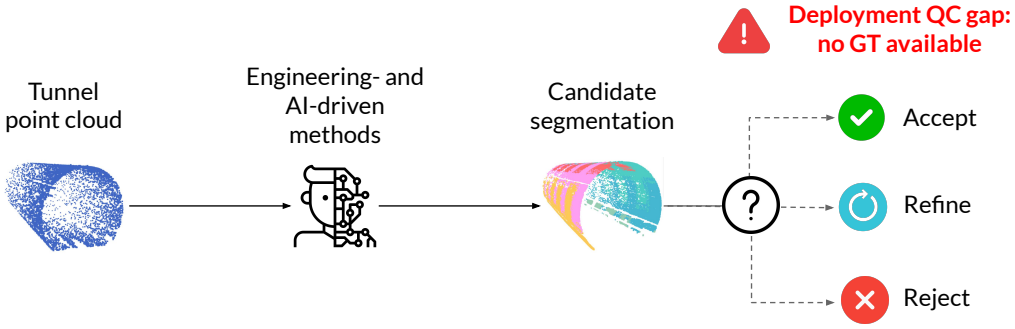


Figure 1: Deployment quality-control (QC) problem: segmentation methods produce candidate outputs, but practitioners still need to decide whether to accept, refine, or flag each result. Without new GT labels, this decision requires other evidence about segmentation accuracy.

Our hypothesis is that parameter changes produce recognisable patterns in these measurements when processing succeeds or fails, and that their relationship with accuracy persists on unseen scans from the same lining family.

Within a pipeline adapted from SAM4Tun [16], we generate calibration runs using one labelled development subset and one expert-tuned reference configuration per lining family (staggered, continuous, and complex). We first sample settings across prescribed parameter ranges, then use uncertainty-guided Bayesian exploration to select further settings for evaluation. All calibration runs within a family reuse the same scan annotations. We test the resulting estimator on held-out scans and deliberately degraded outputs, then use it to guide output selection in a bounded refinement loop. Three separately evaluated LLM proposal systems and an adaptive random-search control generate parameter changes under the same selection rule. The approach is deployment-label-free: segmentation GT labels are used only during development and offline evaluation, and are excluded from scoring, proposal guidelines (e.g, LLM prompts), and parameter selection.

Our experiments also show that proxy scores do not always closely correspond to GT mIoU. We therefore position Proxy4Tun as an exploratory QC framework rather than a deployment-ready tool for verifying segmentation accuracy.

This paper makes the following contributions:

1. **Deployment-label-free QC framework.** We develop an estimator of tunnel-lining segmentation accuracy from PQ and SC measurements using limited development labels, then freeze it before held-out evaluation.
2. **Empirical evaluation of proxy transfer and discrimination.** We assess which measurements transfer across lining families and whether the frozen estimator distinguishes reference outputs from deliberately degraded counterparts on scans excluded from fitting.
3. **Proxy-guided refinement and retrospective analysis.** We evaluate GT-free refinement with LLM and adaptive random-search under the same proxy. Gains concentrate at initial GT mIoU of 0.5–0.7 and mainly arise from reducing whole-ring class swaps.

The rest of this paper is organised as follows. Section 2 reviews related work. Section 3 presents the methodology and experimental design. Section 4 reports the results, Section 5 discusses findings and limitations, and Section 6 concludes.

2. Related work

2.1. Segmentation, adaptation, and quality estimation

Tunnel point-cloud segmentation methods differ in how they use geometric knowledge and labelled data. Engineer-designed pipelines use geometric rules and thresholds [4, 6, 7]; general supervised point-cloud architectures such as PointNet++, RandLA-Net, Mask3D, and TD3D learn features from labelled clouds [17, 18, 19, 20]; tunnel-specific networks such as LiningNet integrate structural geometry [21]. Labelled tunnel data remain limited, and adapting supervised models may require additional annotations [8, 9].

Image-based foundation models such as SAM, SAM2, and SEEM can reduce task-specific training in pipelines that project point clouds into 2D representations [11, 22, 23], with related infrastructure applications including crack segmentation and Scan-to-BIM [24, 12, 25, 26, 27]. SAM4Tun unfolds linings into panoramas and prompts SAM without task-specific training [16], but its preprocessing and prompting still depend on geometry- and scan-specific parameters. R4Tun uses reference information and an LLM to propose bounded adjustments without GT access [28], leaving open how to assess initial and adjusted outputs when new labels are unavailable.

Quality estimators predict segmentation accuracy from observable properties, with examples in medical imaging [29, 14] and driving scenes through LidarMetaSeg [30]. Consistency checks provide another form of evidence, but agreement does not necessarily establish correctness because outputs can share the same errors [31]. Reference-based approaches such as RCA and ConfIC-RCA avoid GT for the new case but still use labelled references during assessment [13, 32]. Proxy4Tun scores each run from its

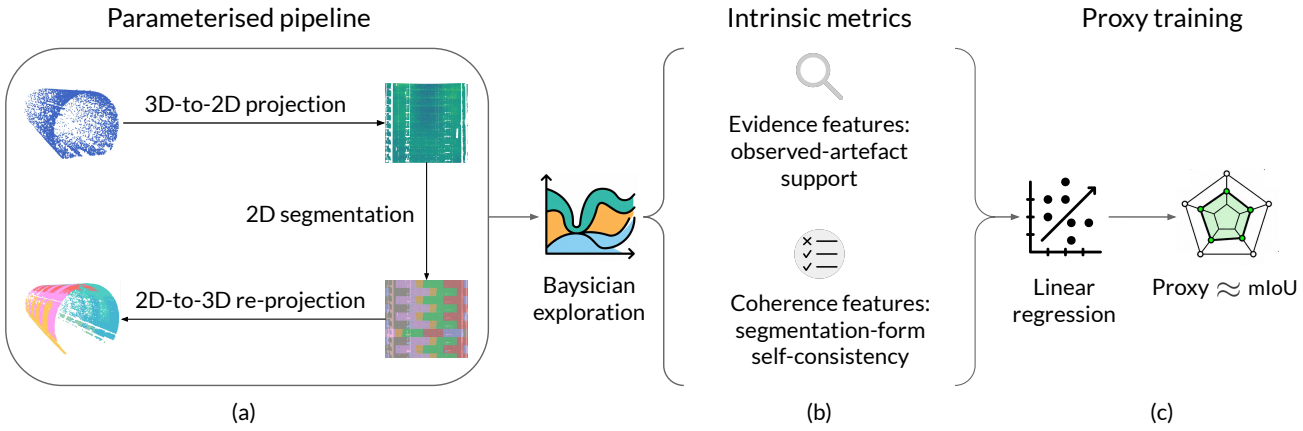


Figure 2: Development and use of the accuracy proxy. Parameter variations generate segmentation outputs with different accuracies. PQ and SC measurements from each run are paired with GT mIoU to train a Ridge proxy. Once frozen, the proxy estimates accuracy on new runs using the four deployment measurements available without GT. The score is an estimate of segmentation accuracy, not a guarantee of correctness.

own PQ and SC measurements using a proxy calibrated beforehand with one labelled subset per lining family.

2.2. Calibration and feedback-guided refinement

Training a quality estimator needs examples spanning successful and degraded segmentations; Robinson et al. generate multiple segmentations by varying model settings and reuse existing annotations [14]. Bayesian optimisation and space-filling designs guide which settings to evaluate [33, 34, 35, 36, 37]. Proxy4Tun combines space-filling sampling with uncertainty-guided Bayesian exploration within prescribed parameter ranges around an expert reference, generating calibration examples that span successful and degraded outcomes.

LLM agents can revise results through critique and feedback [38, 39, 40, 41], but self-correction without external feedback can fail and an imperfect proxy can be over-optimised [42, 43]. Feedback is more useful when it describes what may be wrong [44]. Proxy4Tun supplies PQ and SC measurements for diagnosis and a compact linear score for comparing candidates [45]. The refinement study examines whether LLM proposals improve outcomes compared with adaptive random search under the same proxy-guided selection rule.

3. Methodology

Proxy4Tun has two phases (Fig. 2). During development, we vary pipeline parameters on labelled scans and learn how PQ and SC measurements relate to segmentation accuracy; the fitted proxy is then frozen. At deployment, the frozen proxy estimates mIoU without new GT labels, admits a run to refinement or not, and ranks the outputs of bounded parameter proposals. We first define the task and pipeline, then describe calibration, the proxy, the refinement loop, and the experimental design.

3.1. Task definition

Given a raw point cloud of a tunnel section, the task is to assign each point a positional lining-block class (key block K, adjacent blocks B1/B2, standard blocks A1–An; Fig. 3) or background. Seg2Tunnel, the benchmark dataset we use, groups linings into three families (staggered, continuous, and complex) [17]. These families differ in ring geometry and acquisition conditions and, most critically for our task, in the key-block layout: ring-to-ring repeating, fixed ring-wise order, and no repeating pattern. A ring holds six blocks in the staggered and continuous families and seven in the complex family.

Proxy4Tun assesses a completed segmentation run. Given its intermediate processing outputs and the expected ring count, the proxy estimates the mIoU that would otherwise require comparison with GT labels. The score supports assessment of the current result and comparison with outputs produced by alternative parameter settings on the same scan. At deployment the proxy remains fixed, and GT labels for the new scan are unavailable to both the scorer and the refinement process.

3.2. Framework overview

The framework connects three components: a parameterised segmentation pipeline, an offline procedure for learning the mIoU proxy, and a deployment-time refinement loop. The pipeline produces both segmentations and the measurements used to assess them. Calibration learns the relationship between these measurements and GT mIoU. Refinement uses the frozen proxy to compare new candidates. The processing algorithms and SAM weights remain fixed throughout the study; exploration and refinement change only permitted parameter values.

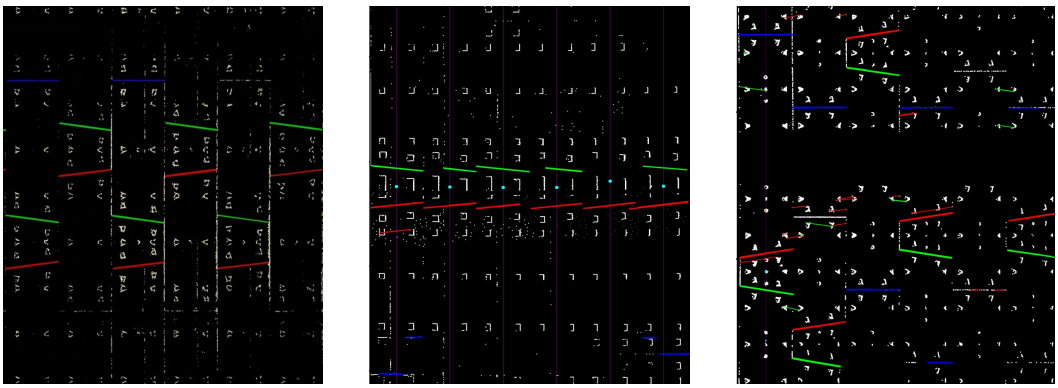


Figure 3: Tunnel-lining block types: key block (K), adjacent block (B), and standard block (A), with background labelled separately. A ring contains six or seven lining blocks depending on its family.

3.2.1. Segmentation pipeline

The pipeline is adapted from SAM4Tun [16] for the three Seg2Tunnel lining families. Figure 4 illustrates the five deployable stages.

(1) *Unfolding*: RANSAC-based ellipse fitting and centreline estimation [18] establish a cylindrical frame in which the lining is represented as a panoramic depth image (radial distance from the centreline). (2) *Denoising*: radial masks and grid-based density filtering [19] remove points inconsistent with the lining surface, trading unwanted-point removal against retained lining data. (3) *Enhancement*: curvature-guided interpolation and progressive upsampling complete the depth image while preserving joints. (4) *Boundary detection*: orientation-specific Hough transforms [20] recover candidate ring and block boundaries, organised by the expected family layout. (5) *Segmentation and reprojection*: the detected layout defines template-based SAM prompts [11], and the masks are mapped back to the point cloud. A GT-dependent evaluation step runs offline only.

Each lining family has an expert reference configuration specifying a complete parameter set. A candidate update is an overlay listing only the parameters to change, applied to a configuration derived from that reference. The reference configurations are released with the code (Appendix A). Appendix C lists the exploration bounds, which also define the permitted refinement search space.

3.2.2. Generating calibration examples

Calibration requires examples linking observable pipeline measurements to segmentation accuracy under successful and degraded parameter settings. We generate these examples by varying pipeline parameters on one labelled development subset per lining family. All runs on that subset are evaluated against the same annotations, so additional runs require computation but no additional scan labelling.

We collect the examples in two stages (Fig. 5a). First, scrambled-Sobol sampling [21] provides 32 valid trials spread across the prescribed parameter ranges. Second, Bayesian exploration adds eight valid trials at settings whose effect on accuracy is most uncertain: a Gaussian-process (GP) model of GT mIoU against parameter settings is fitted to the completed trials, and the candidate with the largest predictive standard deviation in a random pool is evaluated next [22, 23]. This acquisition explores uncertainty in the accuracy response rather than maximising accuracy, so the records deliberately include poor outputs (GT mIoU from about 0.03 upward). A trial is valid when all stages complete and all ten candidate measurements are finite (121 attempted, one failed). Each family contributes 40 valid runs, giving 120 records, each storing its parameter setting, PQ and SC measurements, and GT mIoU. The GP guides exploration only; the proxy is fitted afterwards from measurements to GT mIoU. Appendix C gives the sampler and GP configuration and the per-family bounds.

3.2.3. PQ and SC measurements

For each completed run we compute ten candidate measurements without using any GT labels (Table 1). PQ measurements describe the completeness and geometric plausibility of the representation supplied to later stages: centreline fit residual, orientation agreement, point retention after denoising, and depth-map completeness after enhancement. SC measurements describe whether the recovered layout is coherent: agreement between detected joints and predicted label boundaries, coverage, label consistency across rings, and the ring-count residual, using an expected count available from design or scan records at deployment without segmentation labels; the residual flags layout-count mismatch as one coherence check alongside correspondence and fill. SC is particularly useful for the complex family because it does not assume a fixed keystone pattern. However, high SC does not guarantee correctness: because joint detections also drive SAM prompting, joint–boundary agreement can share correlated errors with the segmentation. It is evidence, not proof, of correctness.

Let Ω be the depth-image grid (5 mm pixels), $V \subseteq \Omega$ the pixels with defined depth, $\mathcal{P}$ the input points, $\ell(p)$ the predicted label of point p (0 for background, including points removed by denoising), and N_{det} the number of ring columns detected in stage 4. The four measurements retained

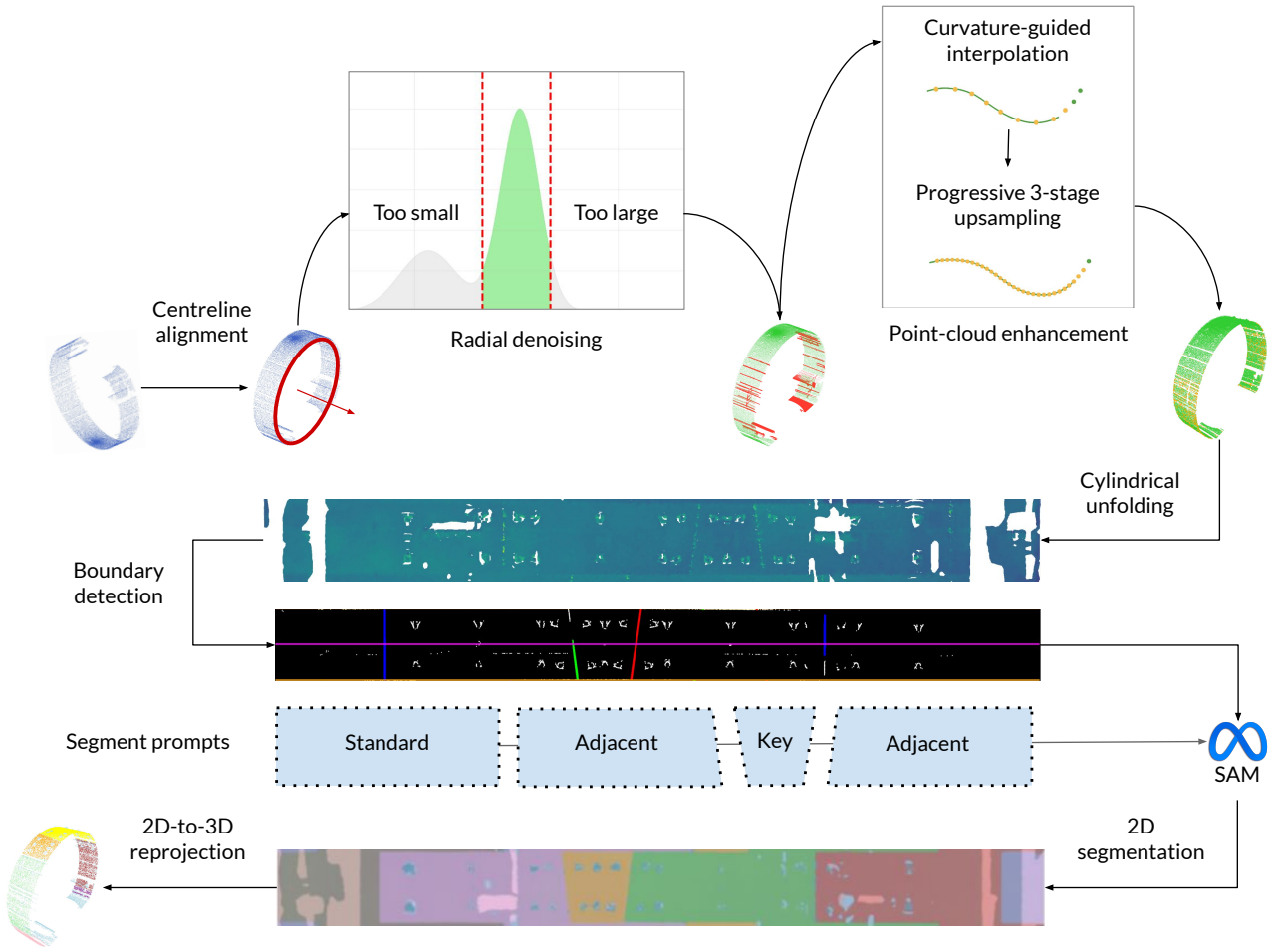


Figure 4: Segmentation pipeline adapted from SAM4Tun. Geometric alignment, filtering, and enhancement produce a depth representation for boundary detection and template-based SAM prompting. The resulting masks are reprojected onto the point cloud. Intermediate outputs provide PQ and SC measurements. The stage algorithms remain fixed; exploration and refinement adjust their parameters.

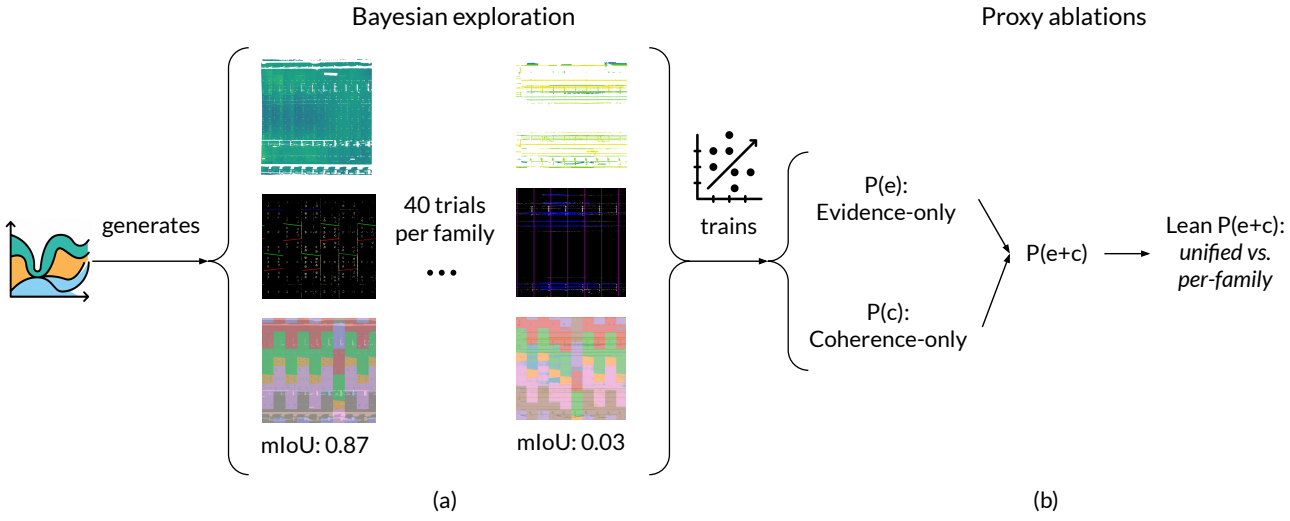


Figure 5: Calibration-example generation and feature comparison. (a) Each lining family contributes 32 valid scrambled-Sobol trials and eight valid Bayesian exploration trials selected by GP uncertainty; thumbnails show three quality levels from one development scan under parameter variation only. (b) LOFO comparison of PQ-only, SC-only, combined (10), and the deployed compact four-feature set (depth NaN ratio, ring-count error, joint correspondence, SAM fill).

Table 1
Candidate measurements for the mIoU accuracy proxy (four PQ, six SC). Pixel quantities refer to the 5 mm depth-image grid. The four retained by the frozen proxy are marked ⋆ and defined in Eqs. (1)–(2). Stages: 1—unfolding; 2—denoising; 3—enhancement; 4—joint detection; 5—SAM segmentation. Combined indices indicate measurements using outputs from both stages.

Group	Measurement	Meaning	Stage
PQ	Unfolding fit residual	Centreline fit error	1
	Orientation agreement	Difference between inferred and expected axes	1
	Point retention	Fraction of input points kept after denoising	2
	Missing-depth rate ⋆	Fraction of depth-image pixels without depth after enhancement	3
SC	Ring-count residual ⋆	$\lvert N_{\text{det}} - N_{\text{exp}}\rvert$ against the design/acquisition prior	4
	Edge support	Fraction of label-boundary pixels within 25 px of a depth edge	3+5
	Correspondence score ⋆	F1 between detected joint lines and predicted block boundaries at 15 px	4+5
	Line–boundary distance	Symmetric mean distance (capped at 200 px) between joints and boundaries	4+5
	Fill rate ⋆	Fraction of input points assigned a non-background label	5
	Phase incoherence	Inconsistency of label ordering across rings	5

by the frozen proxy (Section 4.1) are

$$\begin{aligned} x_{\text{nan}} &= \frac{|\Omega \setminus V|}{|\Omega|}, \\ x_{\text{ringerr}} &= |N_{\text{det}} - N_{\text{exp}}|, \\ x_{\text{fill}} &= \frac{|\{p \in \mathcal{P} : \ell(p) > 0\}|}{|\mathcal{P}|}, \end{aligned} \tag{1}$$

and the correspondence score x_{corr} measures agreement between rasterised detected joint lines L and boundary pixels B, where different non-background labels meet circumferentially within a predicted ring.

Let $d(u, S)$ denote the shortest distance from pixel u to any pixel in set S. With a matching tolerance of $\delta = 15$ px (75 mm), precision π_δ is the fraction of detected joint pixels lying near a predicted boundary, and recall γ_δ is the fraction of predicted boundary pixels lying near a detected joint. Their harmonic mean gives the F1 correspondence score:

$$\begin{aligned} \pi_\delta &= \frac{|\{u \in L : d(u, B) \le \delta\}|}{|L|}, \\ \gamma_\delta &= \frac{|\{u \in B : d(u, L) \le \delta\}|}{|B|}, \\ x_{\text{corr}} &= \frac{2\pi_\delta \gamma_\delta}{\pi_\delta + \gamma_\delta}. \end{aligned} \tag{2}$$

The score ranges from 0 for no agreement to 1 for complete agreement within the tolerance. We assign $x_{\text{corr}} = 0$ if either set is empty or both precision and recall are zero. Changing the tolerance to 8 or 25 px changes the leave-one-family-out (LOFO) correlation by less than 0.01 (Appendix B).

3.2.4. Learning the accuracy proxy

The proxy maps PQ and SC measurements to an estimated mIoU. We use Ridge regression to obtain a compact model with an explicit scoring rule. Regularisation stabilises estimation under correlated predictors [24, 25], and the linear form makes each measurement's contribution directly interpretable. For a feature vector $\mathbf{x} \in \mathbb{R}^d$, the prediction is

$$\hat{y}(\mathbf{x}) = b + \sum_{j=1}^{d} w_j \frac{x_j - \mu_j}{s_j}, \tag{3}$$

where μ_j and s_j are the training mean and standard deviation, w_j the fitted coefficient, and b the intercept; the penalty α is chosen from 25 log-spaced values in $[10^{-3}, 10^3]$ by leave-one-out cross-validation on the fitting data, and predictions are clipped to $[0, 1]$.

We compare four feature sets: PQ only, $P(q)$; SC only, $P(c)$; all ten measurements, $P(q{+}c)$; and a compact subset of $P(q{+}c)$ (Fig. 5b). Leave-one-family-out (LOFO) validation assesses transfer: models are fitted on two families (80 runs) and evaluated on the third (40), with scaling and penalty chosen inside the training split. The compact set was selected among eight nested candidates (top-k features by standardised coefficient, $k = 3, \ldots, 10$, plus the set of coefficients above 10% of the largest) by LOFO Spearman correlation. Because held-out folds were used to choose the set, LOFO figures are model-selection diagnostics; the frozen 27-subset test of Section 3.4.2 is the unbiased estimate. The selected model was refitted on all 120 records and frozen before any evaluation-panel scoring (Appendix B). A permutation control that shuffles GT mIoU within family (20 repeats) checks that the association is not driven by family means. Regression coefficients describe fitted associations with accuracy, not causal prescriptions for parameter changes. *We evaluate the compact linear proxy as an interpretable scoring rule that makes each measurement's contribution explicit and supports inspection of the feedback used in LLM-guided refinement, without claiming superiority over nonlinear alternatives.*

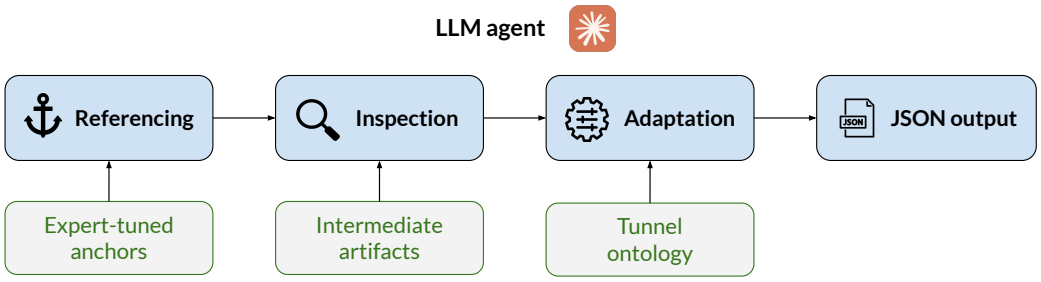


Figure 6: LLM-guided parameter proposals. The agent compares the current run with the family reference, inspects intermediate outputs and PQ/SC measurements, and returns a bounded parameter overlay with a rationale. After execution, the frozen proxy scores the candidate. Candidate retention depends only on the proxy score and a GT-free stage-1 tie-break.

3.3. Proxy-guided parameter refinement

The proxy estimates accuracy but does not determine which parameters should change. A *proposer* supplies this step, and the loop is proposer-agnostic. In this study the proposer is either an LLM agent (Fig. 6) that receives the current settings, intermediate outputs, the four proxy measurements and previous proxy feedback, the family reference, and a structured engineering ontology of failure modes, permitted changes, and stage dependencies (Appendix D), or a sparsity-matched adaptive random sampler that proposes bounded parameter changes around the best configuration found so far, updating this configuration whenever the proxy score improves.

The LLM agent follows an inspect/compare/update loop by chain-of-thought prompting [26, 27]: it grounds itself in the current evidence, links observed symptoms to failure modes, and proposes a small bounded edit (*symptom* → *evidence* → *bounded parameter diff* → *rationale*). A worked three-round trace is given in Appendix D (Table 15).

Each attempt (Fig. 7) obtains one overlay from the proposer, validates it against the family search space (parameter names, types, marginal bounds), executes the valid candidate through stages 1–5, and records its measurements, frozen-proxy score, and final selection. Each admitted case receives three attempts without early stopping; invalid or failed attempts consume an attempt and are excluded from selection.

Admission. The starting proxy score determines admission before the loop, using the study operating bands:

$$\mathrm{QC}(\hat{y}) = \begin{cases} \text{flag}, & \hat{y} < 0.5, \\ \text{refine}, & 0.5 \le \hat{y} \le 0.7, \\ \text{accept}, & \hat{y} > 0.7. \end{cases} \tag{4}$$

which illustrate assessment decisions and are not validated engineering acceptance criteria. Only "refine" starts enter the loop. The same band is applied to the retained output afterwards; a result may remain in the refine band after the fixed budget is exhausted.

Selection. The initial output remains available throughout refinement. If no candidate has a higher proxy score, we retain it. Otherwise, we consider candidates that improve on the initial score.

3.4. Experimental design

The evaluation addresses three questions: whether PQ and SC support accuracy estimation that transfers across lining families, whether the frozen proxy distinguishes reference outputs from controlled degradations on unseen scans, and whether proxy-based selection produces useful parameter refinement, and if so, how much of that effect is attributable to the proposer.

3.4.1. Dataset and use of annotations

We use Seg2Tunnel subsets from three lining families [17] (Table 2). Subsets 2-1, 3-1, and 5-1 are the development subsets and supply the 120 calibration records. The evaluation panel comprises 27 other subsets, nine per family, all excluded from proxy fitting. These subsets come from the represented families and tunnel collection, so the evaluation examines transfer to unseen scans within this setting rather than unrestricted transfer to new projects or lining geometries.

"Held out" refers to proxy calibration only. Table 2 lists every other use of annotation-derived information: the references for 1-1 and 4-1 were tuned on those scans' own GT, and two degraded overlays were identified by GT from earlier trials on evaluation subsets (Section 3.4.2). Evaluation labels are otherwise used only to measure performance and to define retrospective GT strata.

3.4.2. Paired degradation test

Each evaluation subset is processed with its unmodified reference configuration and with a degraded counterpart, giving 27 pairs and 54 runs. The degradation rule was fixed before the proxy was frozen: for each family, the lowest-mIoU completed trial in an earlier parameter-exploration archive supplies a stage-2–5 overlay (Appendix A), applied unchanged on top of every subset's reference with stage-1 settings inherited so that both members share a frame. No label or boundary is edited directly. The test asks whether the frozen proxy ranks the reference above its degraded counterpart and whether the fixed threshold detects the degradation. Because one overlay per family is reused on nine scans, the degraded runs represent three severe parameter failure modes (GT mIoU 0.03–0.12), not 27 independent or marginal failures; the panel supports conclusions about

Table 2
Development and evaluation subsets, and every use of annotation-derived information. One labelled development subset per family supplies the calibration trials; nine other subsets per family form the evaluation panel, all excluded from proxy fitting.

Property	Staggered (T1, T2)	Continuous (T3)	Complex (T4, T5)
Inner diameter / ring length	5.5 m / 1.2 m	5.9 m / 1.2 m	7.5 m / 1.8 m
Blocks per ring; key layout	6; ring-to-ring repeating	6; fixed ring-wise order	7; no repeating pattern
Scanning	Single-station	Multi-station	Single-station, offset
Development subset (40 runs)	2-1	3-1	5-1
Evaluation subsets	1-1–1-5; 2-2–2-5	3-2–3-10	4-1–4-5; 5-2–5-5
Reference source per tunnel	1-1 (own GT); 2-1	3-1 profile	4-1 (own GT); 5-1
Degraded-overlay source (GT mIoU)	2-1 trial (0.032)	3-2 trial (0.027)	4-1 trial (0.036)
Additional uses: every stage-1 run reads the ring-index annotation for frame orientation and validation.			

discrimination under these perturbations rather than operational false-alarm rates.

3.4.3. Refinement panel and proposal arms

The starting run of every evaluation subset is its reference run, shared with the degradation test. The 22 starts with $0.5 \leq \hat{y}_0 < 0.7$ form the refinement panel; three starts fall below and two above the band, and GT plays no part in admission. Each case receives three proposals from each of three LLM proposal systems, Fable 5.1, GPT-5.6, and Gemini 3.8, and from a fourth *random-overlay* control arm, under an identical proxy, validator, budget, execution code, and selector (Fig. 7; Appendix D). All 264 proposals ($22 \times 3 \times 4$) executed. Stage 1 was unlocked in four cases (3-4, 3-8, 3-10, 4-3).

The LLM arms are *session-attributed proposal systems*, not verified model interventions: identifiers come from agent-interface sessions rather than provider API records, decoding settings were not controllable, and each arm used a fresh per-case context resumed only for rounds 2–3 of that case (for Gemini 3.8 most cases had all three proposals prepared before execution). The LLM comparison is therefore exploratory and compares realised proposal systems, not intrinsic LLM capability.

The random-overlay arm asks whether an LLM is needed at all as the proposer. For each case it draws three overlays non-adaptively with a fixed seed: the number of changed parameters k is sampled from the pooled empirical distribution of parameters the three LLM arms actually changed on the 198 panel records (excluding $k = 0$), and the chosen parameters receive values drawn uniformly within the family bounds, with unfolding keys available only when stage 1 is unlocked. The control matches proposal sparsity, not proposal geometry or step size.

To separate the selector from the proposals, each arm is also summarised under reference policies applied to the same executed candidates: retain the start; always take round 1; a uniformly random round (reported as the mean of the three rounds); pure maximum-proxy selection with the start eligible; and a GT oracle (best of start and three rounds). These are post-hoc fixed-pool comparisons; an adaptive proposer might have generated different later proposals under a different rule.

3.4.4. Evaluation metrics and uncertainty

Offline segmentation accuracy is point-level semantic mIoU over the whole subset, pooled per class over background plus the positional block classes ($C = 7$ for staggered and continuous, 8 for complex; all classes occur in every subset). Points removed by denoising or outside every SAM mask count as background, so lost lining points are false negatives. Proxy fidelity is reported as MAE against GT mIoU and Spearman ρ; for the paired test, pair-ranking accuracy is the fraction of the 27 pairs in which the reference scores strictly higher than its degraded counterpart, and alarm precision and recall are calculated at the fixed threshold of 0.5 against the reference/degraded construction labels. The primary interval for ρ is a percentile bootstrap over subsets within family (pairs kept together; 2000 replicates); because subsets of one parent tunnel are dependent, a sensitivity interval enumerates all 5^5 ordered resamples of the five parent tunnels.

For refinement, the case effect Δ_i is the GT mIoU of the selected output minus that of the start. We report mean and median Δ, improved/unchanged/harmed counts at three-decimal precision, worst harm, family and GT-stratum means, and 95% intervals from a paired parent-tunnel cluster bootstrap (five clusters; 100,000 resamples). With five clusters these intervals are coarse, and with one realised trajectory per case and arm they exclude proposer stochasticity. Efficiency is described by proposal yield (candidates whose proxy exceeds the start), accepted-proposal precision (the fraction of those whose GT mIoU also exceeds the start), and selector regret (best available minus selected GT mIoU); LLM latency and cost are not compared because commensurate metadata are unavailable.

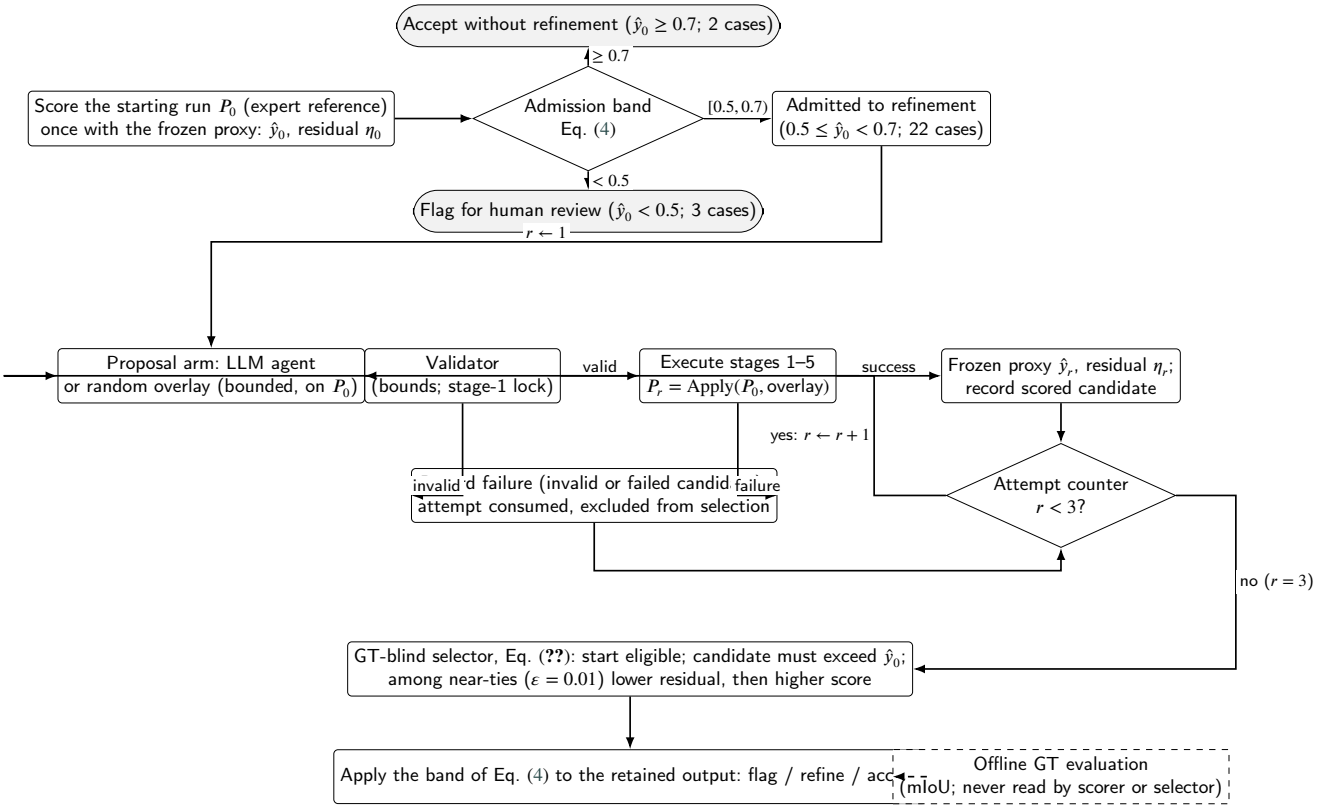


Figure 7: Refinement protocol. Top: the frozen proxy scores the starting run once and the band admits, accepts, or flags it. Middle: each of three attempts yields one bounded overlay from the proposal arm (an LLM agent or the random-overlay control) that is validated, executed, and scored; failures consume an attempt. Bottom: the GT-blind selector chooses among the start and scored candidates, and the band is applied to the retained output. GT is read only by the offline evaluator.

4. Results

The results follow the three questions of Section 3.4: Section 4.1 compares feature sets on the calibration runs; Section 4.2 tests the frozen proxy on the 54 stress-test runs; Sections 4.3–4.4 report the four-arm refinement panel and separate the selector from the proposers; and Section 4.5 analyses where the proxy, the segmentation, and the selector err.

4.1. Proxy calibration and transfer between families

Table 3 and Fig. 8a compare feature sets on the 120 development runs. The ten-feature model fits the calibration data most closely but transfers worst among the combined models; the four-feature compact model has the highest LOFO correlation (0.828) and lowest LOFO MAE (0.112). SC alone transfers better than PQ alone, and the compact combination of one PQ and three SC measurements transfers better than either group. The folds differ substantially (held-out continuous $\rho = 0.627$ against 0.942 staggered and 0.915 complex), so transfer to the multi-station continuous family is the weakest link in the calibration evidence. Within-family label permutation collapses ρ to about 0.2, so the association is not an artefact of family means. Having been used to choose the feature set, these figures are diagnostics rather than unbiased transfer estimates.

The selected proxy contains one PQ and three SC measurements: missing-depth rate, correspondence F1@15, fill rate, and ring-count residual. With standardised inputs z_j, the frozen proxy is

$$\hat{y} = 0.350 + 0.0823 z_{\mathrm{corr}} + 0.0798 z_{\mathrm{fill}} - 0.0454 z_{\mathrm{ringerr}} - 0.0644 z_{\mathrm{nan}}, \tag{5}$$

with $\alpha = 10$ and constants in Appendix B. Higher scores go to segmentations that are populated and supported by detected joint evidence; lower scores to implausible ring counts and incomplete depth maps. These are conditional associations in a correlated linear model, not instructions for changing parameters.

4.2. Paired degradation test

On the 54 evaluation runs the frozen proxy reaches $\rho = 0.817$ and MAE 0.109 (Table 4; Fig. 8b). It ranks the reference above its degraded counterpart in all 27 pairs and, at the 0.5 threshold, flags all 27 degraded runs and three reference runs (3-2, 3-5, and 4-4, whose GT mIoU is 0.631, 0.588, and 0.347). The two interval constructions agree on the direction of the result but differ in width: resampling subsets gives $[0.761, 0.864]$, resampling parent

Table 3
Feature-set comparison on the 120 development runs. LOFO: leave-one-family-out, mean of three folds. Per-fold LOFO for the deployed model: staggered $\rho = 0.942$, MAE 0.122; continuous 0.627, 0.125; complex 0.915, 0.088. Permutation control (deployed set, 20 within-family shuffles): $\rho = 0.202 \pm 0.089$, MAE 0.196 ± 0.008.

		Training fit		LOFO	
Feature set	#Features	ρ	MAE	ρ	MAE
PQ only, $P(q)$	4	0.785	0.108	0.597	0.155
SC only, $P(c)$	6	0.897	0.086	0.766	0.139
Combined, $P(q+c)$	10	0.915	0.081	0.794	0.197
Compact $P(q+c)$ (deployed)	**4**	0.904	0.086	**0.828**	**0.112**

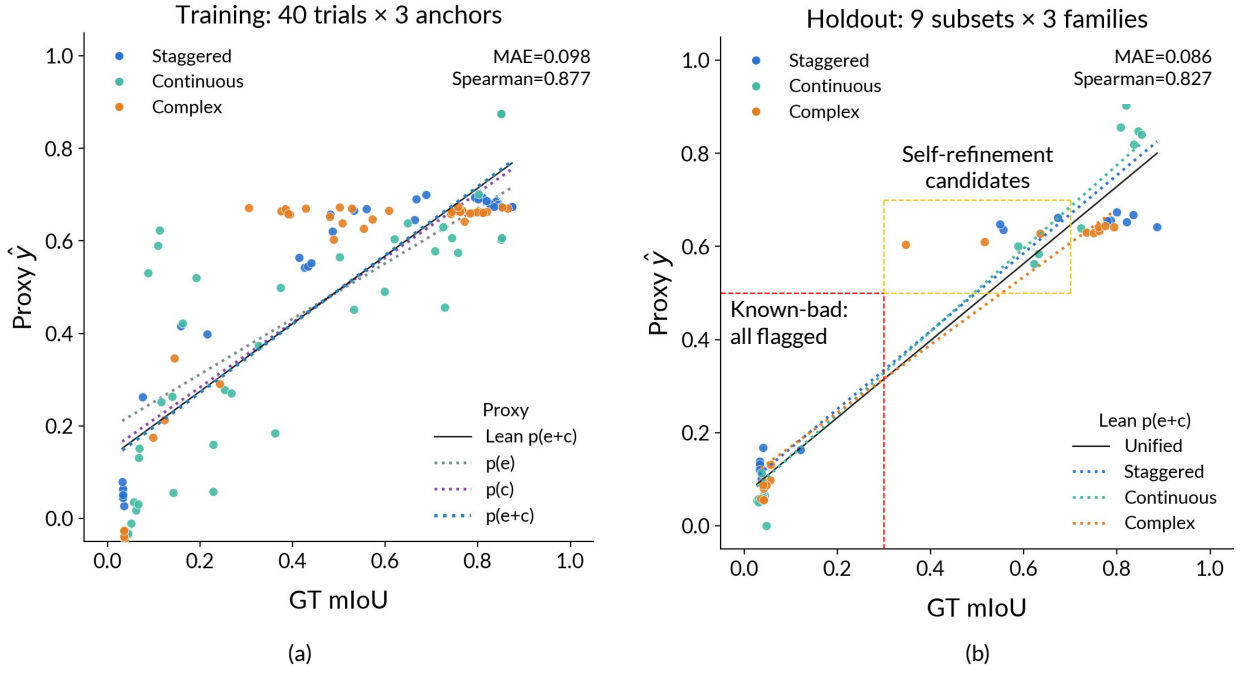


Figure 8: Proxy calibration and frozen evaluation. (a) On the 120 calibration trials (40 per development subset), the compact four-feature model preserves training fit close to the ten-feature model while transferring better between families (Table 3); trend lines show the fitted association of each feature set. (b) The frozen proxy is then evaluated, without refitting, on 54 runs from 27 held-out subsets: 27 reference runs (circles) and their 27 degraded counterparts (crosses). The dashed line is the alarm threshold $\hat{y} < 0.5$; the shaded band $[0.5, 0.7)$ is the refine band from which the 22 refinement cases are admitted. GT is used only on the horizontal axis.

tunnels $[0.718, 0.921]$, so the dependence structure matters at the second decimal.

Ranking and numerical fidelity diverge. An MAE of 0.109 is about half the width of the 0.2-wide refine band, and the proxy under-estimates accurate outputs: the 22 admitted starts have mean proxy 0.606 against mean GT mIoU 0.738, which is why 16 of the 22 "refine" cases were already above 0.7 GT mIoU. Perfect pairwise ranking of severe degradations therefore coexists with absolute estimates too coarse for acceptance decisions.

4.3. Proxy-guided refinement with four proposal arms

From a common start, selected mean mIoU rises to 0.751 (Fable 5.1), 0.760 (GPT-5.6), 0.759 (Gemini 3.8), and 0.752 (random); every pairwise cluster-bootstrap interval among the four arms includes zero (Table 5; Fig. 9a). Median gains are 0.000: the means are driven by a few large improvements. Ranking by mean is not ranking by safety—worst case losses span −0.195 (Fable 5.1) to −0.117 (random).

Table 6 and Fig. 9b isolate the selector on the same candidates. Unselected proposals harm three of four arms on average; the monotone-accept selector recovers 47–78% of the GT-oracle gain. Max-proxy matches monotone-accept for almost every case–arm pair (one Gemini 3.8 tie-break on 3-4). The collapse and recovery of unselected adaptive random search is the strongest within-study evidence that proxy ranking with the start eligible, not proposer identity, carries the gain. Proposal yield differs by arm without changing this picture: of 66 candidates each, Fable 5.1, GPT-5.6,

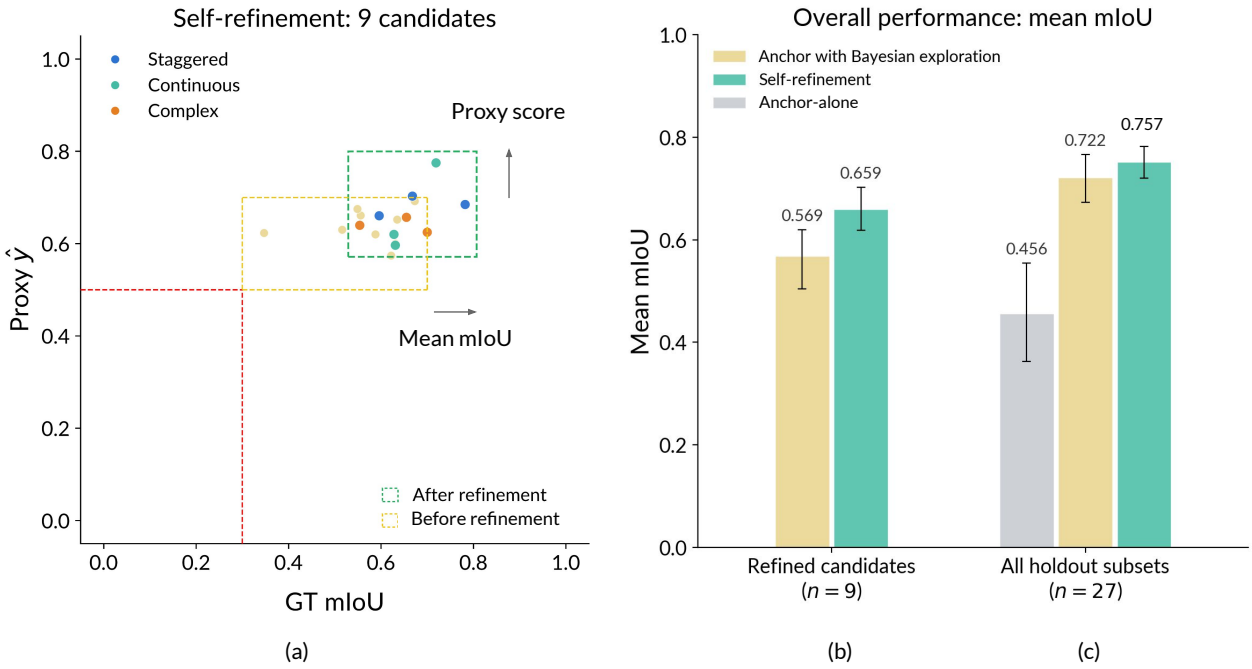


Figure 9: Proxy-guided refinement on the 22 proxy-admitted cases. (a) Mean GT mIoU of the selected output by proposal arm; error bars are 95% parent-tunnel cluster-bootstrap intervals on the paired gain, drawn around the arm mean. (b) The same executed candidates under four selection policies (Table 6): unselected proposals (random round, round 1) fall below the start for three of four arms, whereas the shared GT-blind proxy selector recovers 47–78% of the GT-oracle gain. (c) Per-case gain against starting GT mIoU for all four arms: gains concentrate on the six starts below 0.7 GT mIoU; the remaining 16 cases change little in either direction.

Table 4
Frozen-proxy fidelity on the 54 stress-test runs (27 reference, 27 degraded). Intervals: subset bootstrap within family (primary) and enumeration of all 5^5 parent-tunnel resamples (sensitivity).

Metric	Value
Spearman ρ	0.817
subset-bootstrap 95% CI	[0.761, 0.864]
parent-cluster interval	[0.718, 0.921]
MAE overall	0.109
MAE stag. / cont. / compl.	0.126 / 0.122 / 0.078
Pairs ranked correctly	27/27
Alarms ($\hat{y} < 0.5$), degraded; reference	27/27; 3/27
Alarm precision / recall	0.90 / 1.00

Gemini 3.8, and random produced 43, 22, 36, and 25 proxy-improving candidates, of which 39–56% also improved GT mIoU, with mean selector regret of 0.014, 0.010, 0.006, and 0.017.

4.4. Where the gains occur

Gains concentrate on staggered tunnels and on the six starts with GT mIoU below 0.7; continuous and already-accurate cases barely move (Table 7; Fig. 9c). Subset 1-5 alone can supply about half of an LLM arm's positive gain. Excluding the two self-tuned references (1-1, 4-1) leaves the panel means essentially unchanged. The complete case table is Appendix E. These GT strata are retrospective and cannot serve as a deployment rule.

4.5. Error analysis

Three questions remain: where the proxy's numerical error comes from, which segmentation errors refinement removes, and why the selector sometimes retains a worse output. Point-level outcomes are classified against GT as *correct*; *false negative* (FN, a lining point predicted as background); *false positive* (FP, a background point predicted as lining); and *class swap* (a lining point assigned to the wrong lining block). Fractions are of all points in a subset (Table 8; Fig. 10).

Proxy error. On the 54 stress-test runs the signed error $\hat{y} - y$ is one-directional within each group: reference runs are under-estimated by 0.124 on average (worst for continuous, −0.198) and degraded runs over-estimated by 0.076, the range compression expected from a regularised linear model. Missing-depth and fill saturate on the degraded runs (0.89 and 0.04 against 0.16 and 0.71 for reference runs) and the ring-count residual is zero in all 54 runs, so the order *within* the reference group rests on correspondence and, to a lesser extent, fill. The three reference alarms (3-2, 3-5, 4-4) are the three reference runs with the lowest correspondence F1 (0.06–0.12), the only term that contributes negatively for them. Two are moderately accurate outputs (GT mIoU 0.631

Table 5
Paired refinement results on the 22-case proxy-admitted panel. 95% CIs: parent-tunnel cluster bootstrap. Up/same/down at three-decimal precision. Pairwise mean differences: Fable−GPT −0.009 [−0.033, +0.025]; Fable−Gemini −0.008 [−0.033, +0.019]; Fable−random −0.001 [−0.038, +0.043]; GPT−Gemini +0.001 [−0.049, +0.048]; GPT−random +0.008 [−0.010, +0.022]; Gemini−random +0.007 [−0.049, +0.066].

Arm	Proxy	mIoU	Mean Δ	Median Δ	95% CI	Refined	Up / same / down	Worst Δ
Start (common)	0.606	0.738	–	–	–	–	–	–
Fable 5.1	0.631	0.751	+0.013	0.000	[−0.017, +0.052]	21/22	9 / 4 / 9	−0.195
GPT-5.6	0.629	0.760	+0.022	0.000	[−0.014, +0.077]	15/22	7 / 7 / 8	−0.091
Gemini 3.8	0.641	0.759	+0.021	+0.003	[+0.009, +0.044]	18/22	12 / 7 / 3	−0.024
Random	0.628	0.752	+0.014	0.000	[−0.028, +0.078]	16/22	8 / 8 / 6	−0.117

Table 6
Mean mIoU on the 22-case panel under alternative selection policies applied to the same executed candidates. All policies except the oracle are GT-blind and deployable; all are evaluated with GT. Random round is reported as its expectation (mean of three rounds). Oracle recovery = (monotone-accept − start)/(oracle − start). Post-hoc fixed-pool comparison.

Arm	Start	Random round	Round 1	Max-proxy	Monotone-accept	GT oracle	Oracle recovery
Fable 5.1	0.738	0.734	0.714	0.751	0.751	0.765	49%
GPT-5.6	0.738	0.706	0.683	0.760	0.760	0.770	69%
Gemini 3.8	0.738	0.715	0.692	0.757	0.759	0.765	78%
Random	0.738	0.535	0.508	0.752	0.752	0.769	47%

and 0.588) on multi-station continuous scans with weak detected joint evidence; the third (4-4, GT mIoU 0.347) is a poor output that the construction labels count as a false alarm.

Segmentation error. The degraded overlays act by removing lining points: 68.6% of all points become false negatives (Fig. 10a), which is why the missing-depth and fill terms detect every one of them. Reference runs err differently (7.1% swaps, 5.2% FP, 3.6% FN), and the error maps show that swaps occur as whole-ring bands (Fig. 10b, d): when the key block of a ring is mislocated, the template labelling inherited from SAM4Tun rotates every block in that ring [28]. Refinement acts on this error type and on little else: over the 22 cases the swap fraction falls from 6.6% to 5.4–5.7% while FN and FP change by at most 0.3 points, and the reduction is concentrated on the staggered family (8.9% at the start, 4.7–6.6% after selection), where the gain in Table 7 also concentrates. Continuous starts lose points instead (6.1% FN) and complex starts over-segment (8.3% FP); neither fraction is reduced by any arm, so the two families with little gain fail in ways that the permitted stage-2–5 changes did not repair within three attempts.

Selector error. Of 264 candidates, 118 scored above their start and 68 of these (57.6%) also improved GT mIoU; of 146 rejected candidates, 17 (11.6%) would have improved GT. Admission precision rises with $\Delta\hat{y}$ (23% for $\leq$ 0.01, 90% above 0.05). Within a case the proxy moves almost only through correspondence. Fig. 10e–f shows a near-flat case on 1-4 under Fable 5.1: rounds 1 and 2 score 0.627 and 0.650 while GT mIoU is 0.536 and 0.557—small proxy gains need not track useful accuracy change when joints stay fixed. A minimum admission margin and a label-order term are candidate remedies, neither evaluated here.

5. Discussion

What Proxy4Tun delivers is a frozen estimate that can compare bounded candidates of the same scan, and a retained-start selector that turns any such proposal pool into a modest mean gain. The qualifications that follow bound that claim; they do not reverse it.

Table 7
Mean ΔmIoU by lining family and by starting GT mIoU (retrospective strata for all arms).

Group	n	Fable 5.1	GPT-5.6	Gemini 3.8	Random
Staggered	7	+0.068	+0.076	+0.045	+0.065
Continuous	7	−0.005	+0.001	+0.011	−0.026
Complex	8	−0.019	−0.007	+0.009	+0.005
Start GT mIoU < 0.7	6	+0.084	+0.102	+0.054	+0.086
Start GT mIoU ≥ 0.7	16	−0.014	−0.008	+0.009	−0.012

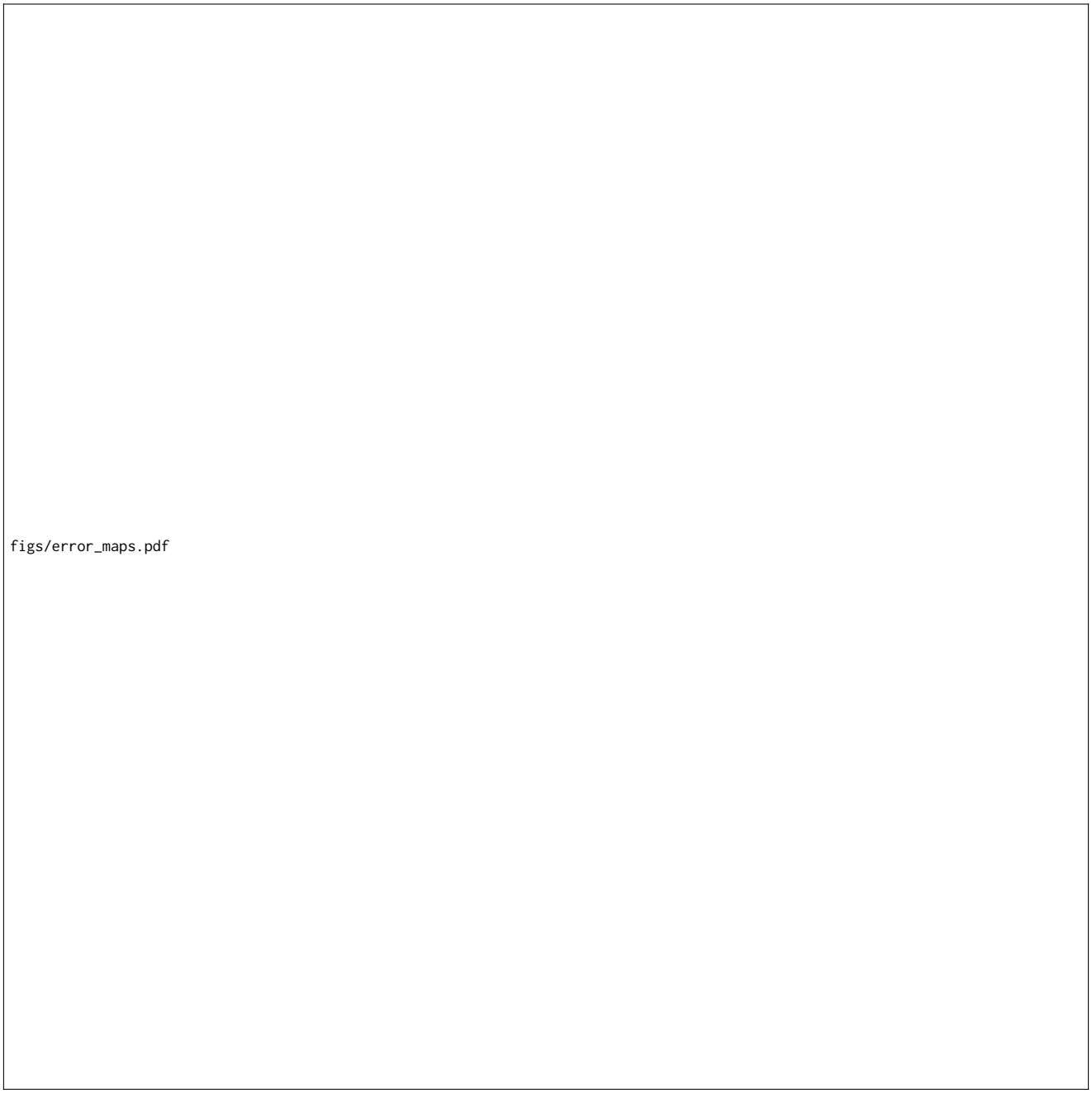


Figure 10: Per-point error maps on the unfolded lining surface (horizontal: circumferential arc position; vertical: axial position; both in metres), classified against GT. Top row, subset 1-5: (a) the degraded overlay of the paired test removes lining points (71% FN) and scores $\hat{y} = 0.145$; (b) the reference run, which is also the refinement start, is dominated by whole-ring class swaps (22%); (c) the Fable 5.1 output retained by the selector (round 2, GT mIoU 0.886) removes most rotated rings (swap 2%). Bottom row, subset 1-4: (d) the start; (e) round 1, GT mIoU 0.536, not selected; (f) round 2, GT mIoU 0.557, selected because its proxy is higher. White regions are scan gaps; cells are coloured by an error type when it covers at least 30% of their points.

The four measurements transfer better as a compact set than as PQ-only, SC-only, or the full ten-feature model, and on scans excluded from fitting they order every reference above its degraded counterpart. That supports *comparison* of outputs of the same scan. Absolute acceptance is not supported: MAE 0.109, systematic under-estimation of accurate outputs, three reference alarms, and the severity of the constructed degradations limit what the 0.5/0.7 bands mean for engineering certification.

Unselected proposals lower the mean for three of four arms—catastrophically for random—while the shared GT-blind selector lifts all four arms and recovers most of the GT-oracle gain. The start is retained when nothing scores higher, so the proxy score is non-decreasing, not the accuracy: small proxy gains can still select whole-ring label rotations to

Table 8
Point-level error composition (% of all points, mean over subsets). Degraded and reference: the 27 stress-test pairs. Start and arms: the 22 admitted cases, with the selected output of each arm. Refinement changes the swap fraction; FN and FP move by at most 0.3 points.

Output	Correct	FN	FP	Swap
Degraded (27)	27.7	68.6	3.2	0.5
Reference (27)	84.2	3.6	5.2	7.1
Start (22)	85.5	2.6	5.3	6.6
Fable 5.1 selected	86.3	2.7	5.3	5.7
GPT-5.6 selected	86.8	2.4	5.5	5.4
Gemini 3.8 selected	86.5	2.5	5.5	5.6
Random selected	86.3	2.7	5.5	5.4

which the four measurements are blind. The LLM arms differ mainly in risk profile; edges over this sparsity-matched random control are small and statistically unresolved. The study therefore does not establish that an LLM proposer is required once a retained-start proxy selector is in place.

Benefit concentrates on the few low-accuracy starts; the accurate majority barely moves, and a single staggered subset can supply about half of an arm's positive gain. The panel mean therefore overstates how widely benefit is distributed. An observable signal that separates an underestimated accurate output from one that can still benefit is the natural next step.

5.1. Comparison with existing paradigms

Comparisons here concern operating points, not accuracy leaderboards (Table 9). Supervised 3D models on Seg2Tunnel report strong in-distribution accuracy [17] but require per-point labels and re-training for each new lining family, and provide no deployment-time gate. SAM4Tun reaches high accuracy on rings tuned case by case [16], but a single fixed configuration transfers imperfectly off the reference scan (mean 0.722 over the 27 evaluation subsets here) and emits no signal when it degrades. Relative to that static pipeline, Proxy4Tun adds an accept/refine/flag gate without deployment labels; relative to one-shot LLM parameter adapters such as R4Tun [28], it adds a calibrated proxy so that refinement can be *assessed and selected* rather than merely proposed, which the random-control arm shows is where most of the benefit lies. In practice, each pipeline release needs one expert reference per lining family and one offline calibration; the same calibration-and-selection pattern can extend to other parameterised infrastructure pipelines, provided the proxy is re-frozen whenever the pipeline or operating population changes.

5.2. Limitations

Claims rest on a single collection, three lining families, five parent tunnels, and one trajectory per case–arm; five limitations bound the study. First, calibration uses parameter variation around one labelled development scan per family within one dataset; transfer depends on how closely a new tunnel matches its reference, and the weak continuous-family fold already shows within-family variation matters. Second, the executed pipeline is annotation-assisted at stage 1; replacing the ring index by a survey direction or geometric convention would require re-establishing the measurements. Third, the compact linear proxy is auditable but cannot represent nonlinear stage interactions or unfamiliar failures, and its four measurements do not register a rotation of block labels within a ring whose joints are correctly placed. Fourth, the alarm and refinement results use constructed degradations and proxy-admitted panels evaluated retrospectively; neither estimates deployment false-alarm rates on naturally degraded runs. Fifth, the LLM comparison is exploratory: identifiers are session-attributed, stochasticity is unmeasured, and the random control matches sparsity but not step size. A confirmatory design needs verified identifiers, repeated seeds, randomised arm order, and external datasets with natural failures.

6. Conclusions

Proxy4Tun freezes a four-feature Ridge proxy from 120 development trials and deploys it as a GT-blind accept/refine/flag signal. On 54 held-out runs it achieves Spearman 0.817 and MAE 0.109, ranks every reference above its degraded counterpart, and flags all 27 degradations (three reference alarms). On 22 proxy-admitted cases, three LLM arms and a random-overlay control sharing one selector raise mean mIoU from 0.738 to 0.751–0.760 (LLMs) and 0.752 (random); unselected adaptive random search collapse to about 0.53, so selection—not proposer identity—carries most of the gain. The gain is fewer whole-ring class swaps; the proxy remains blind to label rotations between correctly placed joints. The study does not claim a superior LLM, LLM necessity over random overlays, or annotation-free execution.

Data availability

The source code, the frozen proxy package (Table 11), the 120-run calibration and 54-run holdout tables, the three degraded overlays, proposal/selection records for all 264 arm proposals plus the 66 fresh Fable 5.1 records, and the evaluation scripts are available at `https://github.com/Tao-Robominds/Proxy4Tun` (tag `v1.0-aic-submission`). Raw Seg2Tunnel scans remain available from the public dataset release; large intermediate artefacts are available on request [17].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI use

Fable 5.1, GPT-5.6, and Gemini 3.8 were evaluated as proposal systems in the refinement experiment. LLMs were

Table 9
Positioning Proxy4Tun against representative tunnel-segmentation paradigms by setup cost and deployment capability. ✓ indicates the requirement (or capability) applies, "–" that it does not. *Deep-learning training* refers to training or fine-tuning a segmentation network; Proxy4Tun fits only a separate linear proxy on offline labelled development data. mIoU figures are *not* measured under comparable settings: supervised and expert-tuned figures are taken from [17] and [16] under their original protocols, whereas the Proxy4Tun figures are from this study on the 27 evaluation subsets: 0.722 for the fixed expert reference, and 0.732 after proxy-admitted refinement of 22 cases with the Fable 5.1 arm (0.739 for GPT-5.6 and Gemini 3.8; 0.733 for random), with the five non-admitted subsets unchanged.

Method	Deep-learning training	Per-case tuning	Deployment-label-free quality signal	mIoU (setting)
LiningNet / SparseUNet / SCF-Net / FA-ResNet	✓	–	–	0.83–0.90 (supervised, in-distribution)
SAM4Tun	–	Expert	–	above 0.90 (per-case tuned rings)
SAM4Tun, fixed family reference	–	–	–	0.722 (27 evaluation subsets)
Proxy4Tun (compact $P(q+c)$)	–	**Bounded, auto**	✓	**0.732–0.739** (27 subsets, by arm)

also used for language editing and manuscript structure. The authors reviewed and edited all content and take full responsibility for the content of the publication.

References

[1] L. Attard, C. J. Debono, G. Valentino, M. D. Castro, Tunnel inspection using photogrammetric techniques and image processing: A review, ISPRS Journal of Photogrammetry and Remote Sensing 144 (2018) 180–188.

[2] R. Montero, J. G. Victores, S. MartÃηnez, A. JardÃşn, C. Balaguer, Past, present and future of robotic tunnel inspection, Automation in Construction 59 (2015) 99–112.

[3] A. Strauss, J. Bien, H. Neuner, C. Harmening, C. Seywald, M. Österreicher, et al., Sensing and monitoring in tunnels: testing and monitoring methods for the assessment of tunnels, Structural Concrete 21 (2020) 1356–1376.

[4] M. Q. Huang, J. Ninić, Q. B. Zhang, BIM, machine learning and computer vision techniques in underground construction: current status and future perspectives, Tunnelling and Underground Space Technology 108 (2021) 103677.

[5] A. SjÃűlander, V. Belloni, A. Ansell, E. NordstrÃűm, Towards automated inspections of tunnels: a review of optical inspections and autonomous assessment of concrete tunnel linings, Sensors 23 (2023) 3189.

[6] L. Weidner, G. Walton, Generalized extraction of bolts, mesh, and rock in tunnel point clouds: a critical comparison of geometric feature-based methods using random forest and neural networks, Remote Sensing 16 (2024) 4466.

[7] Z. Wang, Z. Zhu, Y. Wu, Q. Hong, D. Jiang, J. Fu, et al., Automated tunnel point cloud segmentation and extraction method, Applied Sciences 15 (2025) 2926.

[8] C. Su, Q. Hu, Z. Yang, R. Huo, A review of deep learning applications in tunneling and underground engineering in china, Applied Sciences 14 (2024) 1720.

[9] Y.-J. Cha, R. Ali, J. Lewis, O. Büyüköztürk, Deep learning-based structural health monitoring, Automation in Construction 161 (2024) 105328.

[10] R. Bommasani, D. A. Hudson, E. Adeli, R. Altman, S. Arora, S. von Arx, et al., On the opportunities and risks of Foundation Models, 2021. URL: https://arxiv.org/abs/2108.07258. doi:10.48550/arXiv.2108.07258. arXiv:2108.07258.

[11] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, et al., Segment anything, in: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2023, pp. 4015–4026. doi:10.1109/ICCV51070.2023.00371.

[12] Z. Ye, L. Lovell, A. Faramarzi, J. Ninić, Sam-based instance segmentation models for the automation of structural damage detection, Advanced Engineering Informatics 62 (2024) 102826.

[13] V. V. Valindria, I. Lavdas, W. Bai, K. Kamnitsas, E. O. Aboagye, A. G. Rockall, D. Rueckert, B. Glocker, Reverse classification accuracy: Predicting segmentation performance in the absence of ground truth, IEEE Transactions on Medical Imaging 36 (2017) 1597–1606.

[14] R. Robinson, O. Oktay, W. Bai, V. V. Valindria, M. M. Sanghvi, N. Aung, J. M. Paiva, F. Zemrak, K. Fung, E. Lukaschuk, A. Lee, V. Carapella, Y. J. Kim, B. Kainz, S. K. Piechnik, S. Neubauer, S. Petersen, C. Page, D. Rueckert, B. Glocker, Real-time prediction of segmentation quality, in: Medical Image Computing and Computer Assisted Intervention – MICCAI 2018, volume 11073 of *Lecture Notes in Computer Science*, 2018, pp. 578–585. doi:10.1007/978-3-030-00937-3_66.

[15] J. Fournel, A. Bartoli, D. Bendahan, M. Guye, et al., Medical image segmentation automatic quality control: A multi-dimensional approach, Medical Image Analysis 74 (2021) 102213.

[16] Z. Ye, W. Lin, A. Faramarzi, X. Xie, J. Ninić, SAM4Tun: no-training model for tunnel lining point cloud component segmentation, Tunnelling and Underground Space Technology 158 (2025) 106401.

[17] W. Lin, B. Sheil, P. Zhang, B. Zhou, C. Wang, X. Xie, Seg2Tunnel: a hierarchical point cloud dataset and benchmarks for segmentation of segmental tunnel linings, Tunnelling and Underground Space Technology 147 (2024) 105735.

[18] M. A. Fischler, R. C. Bolles, Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography, Communications of the ACM 24 (1981) 381–395.

[19] M. Ester, H.-P. Kriegel, J. Sander, X. Xu, A density-based algorithm for discovering clusters in large spatial databases with noise, in: Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96), AAAI Press, 1996, pp. 226–231.

[20] R. O. Duda, P. E. Hart, Use of the Hough transformation to detect lines and curves in pictures, Communications of the ACM 15 (1972) 11–15.

[21] S. Joe, F. Y. Kuo, Constructing Sobol sequences with better two-dimensional projections, SIAM Journal on Scientific Computing 30 (2008) 2635–2654.

[22] R. B. Gramacy, H. K. H. Lee, Adaptive design and analysis of supercomputer experiments, Technometrics 51 (2009) 130–145.

[23] R. B. Gramacy, Surrogates: Gaussian Process Modeling, Design and Optimization for the Applied Sciences, Chapman & Hall/CRC, Boca Raton, Florida, 2020. URL: https://bookdown.org/rbg/surrogates/.
[24] A. E. Hoerl, R. W. Kennard, Ridge regression: Biased estimation for nonorthogonal problems, Technometrics 12 (1970) 55–67.
[25] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, É. Duchesnay, Scikit-learn: Machine learning in python, Journal of Machine Learning Research 12 (2011) 2825–2830.
[26] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, et al., Chain-of-thought prompting elicits reasoning in large language models, in: Advances in Neural Information Processing Systems 35 (NeurIPS 2022), 2022, pp. 24824–24837. doi:10.52202/068431-1800.
[27] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, Y. Cao, React: Synergizing reasoning and acting in language models, 2023. URL: https://arxiv.org/abs/2210.03629. doi:10.48550/arXiv.2210.03629. arXiv:2210.03629, international Conference on Learning Representations (ICLR).
[28] X. Tao, Z. Ye, G. Wang, J. Ninić, B. Sheil, R4Tun: LLM-guided adaptive segmental tunnel lining segmentation in point clouds, 2026. Manuscript.

A. Reference configurations and coordinate conventions

The expert reference configuration of each family is released as a JSON parameter snapshot with the code (Data availability); the snapshots cover the unfolding geometry (diameter 5.5 / 5.9 / 7.5 m and ring length 1.2 / 1.2 / 1.8 m for staggered / continuous / complex), the radial denoising gates, the enhancement thresholds, the Hough accumulator settings, and the SAM template geometry (segments per ring, segment order, key-block height and joint inclination). The continuous-family reference is a profile established on development scan 3-1 and applied unchanged to 3-2–3-10; the other references (1-1, 2-1, 4-1, 5-1) name both scan and profile, and 1-1 and 4-1 were tuned on their own GT. Table 10 lists the three degraded overlays.

Coordinates and units. Stage 1 assigns each point to its nearest sample of the fitted centreline and expresses it by radius r, axial position h, and circumferential arc s measured from the invert, with the centreline direction oriented so that h correlates with the Seg2Tunnel ring-index column ($|\rho_P| > 0.5$, prescribed sign per family). Depth images use 5 mm/px (15 px = 75 mm); `processing.y_bounds` is in millimetres, while `y_step` and `mask_theta_*` are in metres.

B. Frozen proxy package

Table 11 gives the frozen scoring rule so that Eq. (5) can be evaluated from raw measurements. The package was fitted on all 120 calibration runs with $\alpha = 10$ (from the 25-point grid) and intercept $b = 0.350225$; scores are clipped to $[0, 1]$. During development the correspondence tolerance was evaluated within the ten-feature model at 8, 15, and 25 px (LOFO ρ 0.789, 0.794, 0.803; MAE 0.198, 0.197, 0.189); 15 px was fixed before feature selection and not re-optimised.

C. Calibration sampling

Table 12 specifies the example-generation procedure of Section 3.2.2; Table 13 lists the per-family bounds, which also define the refinement validator's search space.

D. Refinement-panel configuration, ontology, and agent interface

Table 14 records the configuration shared by the four arms and the provenance of each arm. The random-overlay arm uses no ontology and no LLM: it draws three non-adaptive overlays per case from a sparsity-matched uniform sampler (seed base 20260915; k from the pooled LLM changed-parameter distribution on 198 panel records).

Ontology and prompt. The LLM proposer consults a staged SAM4Tun ontology (pipeline artifacts, per-stage parameters and quality metrics, failure modes, diagnostic rules, remediation actions, inter-stage dependencies; the YAML is part of the release). Each attempt's prompt is assembled from a fixed skeleton (role, isolation rules, current state, GT-blind measurements and proxy features, allowed artifact images, family bounds) and requires the JSON schema of Listing 1. Full per-attempt prompts are part of the execution records; GT mIoU and evaluation directories are never included.

Listing 1: Required proposal schema (per attempt).

```
{
  "observation": "...",
  "rule": "...",
  "failure_mode": "...",
  "rationale": "...",
  "overlay": { "denoising": {}, "enhancing": {}, "detecting": {}, "
       sam": {} },
  "full": false
}
```

Superseded Fable arm. An earlier Fable 5.1 arm run in a persistent session with pilot exposure gave mean selected mIoU 0.768 ($\Delta = +0.030$) on the same 22-case panel; it is superseded by the fresh per-case Fable 5.1 arm reported in the main text and is not used in any comparison.

Worked example. Table 15 traces subset 1-5 under the GPT-5.6 arm (stage 1 locked, residual 2.5 cm in all rounds). The start scores $\hat{y}_0 = 0.555$ with correspondence F1 0.201. Round 1 loosened the detector to recover "missing and fragmented staggered seams" and scored 0.528, below the start. Reading that feedback, round 2 "deliberately moves away from Round 1's permissive detector" and tightened the same parameters (score 0.571); round 3 tightened further (0.610) and was retained; its offline GT mIoU is 0.887 against 0.549 at the start, the largest gain in the panel. Fill and missing-depth barely change across the rounds: within a case the proxy moves through the correspondence term (Section 4.5). GT values were computed offline and were never available to the agent.

Table 10
Degraded overlays of the paired degradation test: the lowest-mIoU completed trial of each family in an earlier exploration archive, applied on top of every evaluation subset's reference with unfolding inherited. Only parameters differing from the reference are listed (rounded; snapshots hold full precision).

Family	Source trial (mIoU)	Overlay
Staggered	2-1-t002 (0.032)	denoising: `mask_r_low` 2.277, `mask_r_high` 2.731, `z_step` 0.00684, `grad_threshold` 0.186; enhancing: `curvature_threshold` 0.00446, `inter_radius` 0.0228; detecting: `binary_threshold` 134, `hough_threshold_oblique` 72, `hough_threshold_horizontal` 77, `hough_threshold_vertical` 571, `maxLineGap_oblique` 49; sam: `processing.y_bounds` [3720, 13100]
Continuous	3-2-t029 (0.027)	denoising: `mask_r_low` 2.801, `mask_r_high` 2.908, `mask_theta_low` 1.835, `mask_theta_high` 15.00; enhancing: `curvature_threshold` 0.00112; detecting: `binary_threshold` 156, `hough_threshold_oblique` 53, `hough_threshold_horizontal` 38, `hough_threshold_vertical` 1445, `maxLineGap_oblique` 30, `pattern_tolerance` 11, `uniform_k_snap` on; sam: `segment_width` 1366, `K_height` 730.0, `angle` 6.76
Complex	4-1-t009 (0.036)	denoising: `mask_r_low` 3.771, `mask_r_high` 3.958; enhancing: `curvature_threshold` 0.00111; detecting: `binary_threshold` 108, `hough_threshold_oblique` 64, `hough_threshold_horizontal` 64, `hough_threshold_vertical` 1107, `maxLineGap_oblique` 63, `maxLineGap_horizontal` 27, `pattern_tolerance` 16; sam: `segment_width` 1732, `K_height` 1443.5, `angle` 8.03

Table 11
Frozen four-feature proxy: feature order, standardisation constants, and Ridge coefficients ($\alpha = 10$, $b = 0.350225$).

Feature j	μ_j	s_j	w_j
`correspondence_f1@15` (x_{corr})	0.168262	0.150653	+0.082313
`sam_fill_rate` (x_{fill})	0.459842	0.294438	+0.079829
`ring_count_error` (x_{ringerr})	0.583333	0.493007	−0.045448
`depth_nan_ratio` (x_{nan})	0.408375	0.310512	−0.064384

E. Case-level refinement results

Table 16 lists the proxy-selected output and offline mIoU change for every panel case. A denotes retention of the start; R1–R3 the selected round. Selected-round distributions (start/R1/R2/R3) are 1/4/10/7 for Fable 5.1, 7/7/3/5 for GPT-5.6, 4/7/5/6 for Gemini 3.8, and 6/7/5/4 for random.

Table 12
Calibration sampling configuration (frozen snapshot).

Setting	Value
Valid trials per family	40 (32 scrambled-Sobol + 8 GP-uncertainty acquisitions), applied as overlays on the family reference
Sobol mapping	Continuous parameters scaled to bounds; integers rounded, Booleans thresholded; inverted radial gates reset to midpoint ∓ 0.02 m
GP	Matérn $\nu = 2.5$ + white noise; `normalize_y=True`; length-scale bounds $(10^{-2}, 10^{2})$, noise $(10^{-6}, 10^{-1})$; `GaussianProcessRegressor`, 3 optimiser restarts; refitted after each valid trial
Acquisition	Maximise predictive standard deviation over a random pool of 2048 candidates
Validity	Stages 1–5 and the offline evaluation step complete, all ten candidate measurements finite; otherwise excluded and topped up (121 attempts, one failure)

Table 13
Per-family search-space bounds. Units: radial gates, `inter_radius`, `slice_spacing_factor`, `mask_theta_low/high` in m; `y_bounds_lower`, `segment_width`, `K_height` in mm; `angle` in degrees; Hough thresholds in votes; `maxLineGap` and `pattern_tolerance` in pixels; `binary_threshold` in grey levels; `ransac_threshold` in the pipeline's internal units. Unfolding keys are locked in refinement unless the start residual exceeds 10 cm.

Parameter	Staggered (2-1)	Continuous (3-1)	Complex (5-1)
`unfolding.ransac_threshold`	$[0.8, 1.2]$	$[0.8, 1.2]$	$[0.8, 1.2]$
`unfolding.slice_spacing_factor`	$[1.05, 1.35]$	$[1.05, 1.35]$	$[1.6, 2.0]$
`unfolding.polynomial_degree`	$\{2, 3\}$	$\{2, 3\}$	$\{2, 3\}$
`unfolding.random_seed`	$\{0, \dots, 7\}$	$\{0, \dots, 7\}$	$\{0, \dots, 7\}$
`denoising.mask_r_low`	$[2.2, 2.5]$	$[2.7, 3.0]$	$[3.5, 3.8]$
`denoising.mask_r_high`	$[2.7, 2.9]$	$[2.9, 3.15]$	$[3.75, 4.05]$
`denoising.z_step`	$[0.001, 0.008]$	–	–
`denoising.grad_threshold`	$[0.1, 0.25]$	–	–
`denoising.mask_theta_low`	–	$[1.0, 2.5]$	–
`denoising.mask_theta_high`	–	$[15, 19]$	–
`enhancing.curvature_threshold`	$[3\times10^{-4}, 8\times10^{-3}]$	$[2\times10^{-4}, 2\times10^{-3}]$	$[2\times10^{-4}, 2\times10^{-3}]$
`enhancing.inter_radius`	$[0.02, 0.08]$	–	–
`detecting.binary_threshold`	$[100, 160]$	$[100, 160]$	$[100, 160]$
`detecting.hough_threshold_oblique`	$[40, 90]$	$[20, 60]$	$[25, 70]$
`detecting.hough_threshold_horizontal`	$[40, 90]$	$[20, 60]$	$[25, 70]$
`detecting.hough_threshold_vertical`	$[400, 800]$	$[800, 2500]$	$[800, 5000]$
`detecting.maxLineGap_oblique`	$[20, 80]$	$[10, 60]$	$[30, 80]$
`detecting.maxLineGap_horizontal`	–	–	$[5, 30]$
`detecting.pattern_tolerance`	–	$[5, 20]$	$[5, 20]$
`detecting.uniform_k_snap`	–	$\{\mathrm{off}, \mathrm{on}\}$	–
`sam.y_bounds_lower`	$[3600, 4400]$	–	–
`sam.segment_width`	–	$[1000, 1400]$	$[1500, 2100]$
`sam.K_height`	–	$[700, 1000]$	$[1000, 1500]$
`sam.angle`	–	$[4, 9]$	$[7, 12]$

Table 14
Refinement-panel configuration and arm provenance.

Setting	Value
Starting run	Unmodified expert reference run of each subset (shared with the degradation test)
Context (LLM arms)	Engineering ontology, stage outputs, four proxy measurements, stage-1 residual, proxy feedback, family bounds
Random control	Non-adaptive; $k \sim$ empirical LLM changed-parameter distribution; values $\sim U(\text{low}, \text{high})$ in family bounds; seed base 20260915
Validator	Family search space (Table 13); stage-1 keys only when the start residual exceeds 10 cm; empty overlays rejected
Budget and stopping	Three attempts per case and arm; no early stopping; all 264 attempts executed ($22 \times 3 \times 4$)
Selection	Eq. (**??**); no separate coherence gate
Pipeline hardware	One NVIDIA RTX 5060 GPU
LLM interface	Cursor agent sessions. Fable 5.1, GPT-5.6, and Gemini 3.8: one fresh sub-agent per case, resumed for rounds 2–3 (14–15 Sep 2026)
Recorded identifiers	Fable 5.1: interface slug `claude-fable-5-1-thinking-high`; GPT-5.6: `gpt-5.6-sol-medium`; Gemini 3.8: `inherit`. None is a provider API version string
Decoding	Provider defaults through the interface; not controllable except the "medium" effort in the GPT-5.6 slug
Stored selections	Each arm holds the 22 panel cases (66 executions). An earlier persistent-session Fable 5.1 arm is superseded and not used in comparisons; legacy GPT-5.6 runs with a different proxy are excluded

Table 15
Worked example: subset 1-5, GPT-5.6 arm. Overlay values are shown for the parameters the agent changed relative to the staggered reference; `curv` = `curvature_threshold`, `r` = `inter_radius`, `bin` = `binary_threshold`, $\mathrm{H}_o/\mathrm{H}_h$ = oblique/horizontal Hough thresholds, `gap` = `maxLineGap_oblique`, y_{lo} = lower bound of `processing.y_bounds` (mm). The four proxy measurements, the clipped score, the selector decision, and the offline GT mIoU are listed per round.

Round	Overlay (changed parameters)	x_{corr}	x_{fill}	x_{ringerr}	x_{nan}	$\hat{y}$	Decision	GT mIoU
Start	staggered reference (curv 0.005, r 0.03, bin 125, $\mathrm{H}_o/\mathrm{H}_h$ 65/65, gap 45, y_{lo} 3800)	0.201	0.729	0	0.118	0.555	admitted (refine)	0.549
R1	curv 0.002, r 0.05, bin 115, $\mathrm{H}_o/\mathrm{H}_h$ 45/45, gap 70, y_{lo} 4000	0.150	0.734	0	0.120	0.528	below start; inadmissible	0.448
R2	curv 0.0065, r 0.03, bin 150, $\mathrm{H}_o/\mathrm{H}_h$ 80/75, gap 30, y_{lo} 3600	0.230	0.729	0	0.117	0.571	admissible	0.677
R3	curv 0.0075, r 0.025, bin 155, $\mathrm{H}_o/\mathrm{H}_h$ 85/70, gap 25, y_{lo} 3700	0.301	0.728	0	0.117	0.610	admissible; **retained**	0.887

Table 16
Case-level selected mIoU and change from the common start for the four proposal arms.

		Fable 5.1		GPT-5.6		Gemini 3.8		Random	
Subset	Start	Selected (round)	Δ	Selected (round)	Δ	Selected (round)	Δ	Selected (round)	Δ
1-1	0.787	0.786 (R2)	−0.001	0.787 (A)	+0.000	0.763 (R3)	−0.024	0.791 (R3)	+0.004
1-2	0.821	0.835 (R2)	+0.014	0.821 (A)	+0.000	0.842 (R3)	+0.021	0.825 (R1)	+0.004
1-3	0.779	0.780 (R3)	+0.001	0.780 (R3)	+0.001	0.867 (R2)	+0.088	0.786 (R1)	+0.007
1-4	0.556	0.557 (R2)	+0.001	0.809 (R2)	+0.253	0.556 (A)	+0.000	0.811 (R2)	+0.255
1-5	0.549	0.886 (R2)	+0.337	0.887 (R3)	+0.338	0.613 (R2)	+0.064	0.867 (R3)	+0.318
2-2	0.673	0.846 (R2)	+0.173	0.672 (R2)	−0.001	0.839 (R2)	+0.166	0.592 (R1)	−0.081
2-4	0.835	0.783 (R3)	−0.052	0.776 (R3)	−0.059	0.835 (R2)	+0.000	0.784 (R3)	−0.051
3-3	0.808	0.812 (R3)	+0.004	0.806 (R1)	−0.002	0.808 (A)	+0.000	0.691 (R2)	−0.117
3-4	0.622	0.601 (R2)	−0.021	0.627 (R3)	+0.005	0.692 (R1)	+0.070	0.622 (A)	+0.000
3-6	0.836	0.820 (R2)	−0.016	0.837 (R3)	+0.001	0.841 (R3)	+0.005	0.805 (R2)	−0.031
3-7	0.845	0.846 (R1)	+0.001	0.845 (A)	+0.000	0.845 (R1)	+0.000	0.814 (R1)	−0.031
3-8	0.723	0.723 (A)	+0.000	0.723 (A)	+0.000	0.728 (R1)	+0.005	0.723 (A)	+0.000
3-9	0.820	0.818 (R2)	−0.002	0.820 (A)	+0.000	0.817 (R1)	−0.003	0.820 (A)	+0.000
3-10	0.853	0.852 (R1)	−0.001	0.853 (A)	+0.000	0.854 (R2)	+0.001	0.853 (A)	+0.000
4-1	0.635	0.650 (R3)	+0.015	0.652 (R2)	+0.017	0.658 (R3)	+0.023	0.660 (R2)	+0.025
4-2	0.734	0.539 (R1)	−0.195	0.643 (R1)	−0.091	0.744 (R3)	+0.010	0.738 (R1)	+0.004
4-3	0.516	0.516 (R2)	+0.000	0.515 (R1)	−0.001	0.516 (A)	+0.000	0.516 (A)	+0.000
4-5	0.750	0.784 (R1)	+0.034	0.777 (R1)	+0.027	0.780 (R1)	+0.030	0.750 (A)	+0.000
5-2	0.793	0.793 (R3)	+0.000	0.792 (R1)	−0.001	0.782 (R3)	−0.011	0.793 (R1)	+0.000
5-3	0.762	0.761 (R2)	−0.001	0.762 (A)	+0.000	0.762 (R1)	+0.000	0.762 (R2)	+0.000
5-4	0.760	0.760 (R3)	+0.000	0.759 (R1)	−0.001	0.760 (A)	+0.000	0.752 (R1)	−0.008
5-5	0.775	0.772 (R3)	−0.003	0.771 (R1)	−0.004	0.793 (R1)	+0.018	0.794 (R3)	+0.019
Mean	0.738	0.751	+0.013	0.760	+0.022	0.759	+0.021	0.752	+0.014